

(The labels 1 and 2 are used here for fermions and bosons, respectively.) For fermion-fermion scattering there are also two second-order diagrams, shown in Figure 6.4. They yield the S -matrix element

$$\begin{aligned}
 S_{\mathbf{p}_1' \sigma_1' n_1' \mathbf{p}_2' \sigma_2' n_2' ; \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} &= (2\pi)^{-6} \sum_{k'l'm'klm} (-i)^2 g_{m'mk'} g_{l'l k} \\
 &\times u_{m'}^*(\mathbf{p}_2' \sigma_2' n_2') u_{l'}^*(\mathbf{p}_1' \sigma_1' n_1') u_m(\mathbf{p}_2 \sigma_2 n_2) u_l(\mathbf{p}_1 \sigma_1 n_1) \\
 &\times \int d^4x \int d^4y e^{-ip_2' \cdot x} e^{-ip_1' \cdot y} e^{ip_2 \cdot x} e^{ip_1 \cdot y} (-i) \Delta_{k'k}(x-y) \\
 &- [1' \rightleftharpoons 2']
 \end{aligned} \tag{6.1.28}$$

with the last term indicating subtraction of the preceding term with interchange of particles 1' and 2' (or equivalently 1 and 2). There are no second-order graphs for boson-boson scattering in this theory; the lowest-order graphs are of fourth order, such as that shown in Figure 6.7. More specific examples of formulas like Eqs. (6.1.27) and (6.1.28) will be given in Section 6.3, after we have had a chance to evaluate the propagators and go over to momentum space.

In the interaction (6.1.18) of the preceding example, the three fields are all different. It is instructive also to look at an example with a trilinear interaction in which all three fields are the same, or at least enter into the interaction in a symmetric way.

Theory II

Now take the interaction density to be a sum of terms that are trilinear in a set of *real* bosonic fields $\phi_\ell(x)$:

$$\mathcal{H}(x) = \frac{1}{3!} \sum_{\ell mn} g_{\ell mn} \phi_\ell(x) \phi_m(x) \phi_n(x) \tag{6.1.29}$$

with $g_{\ell mn}$ a real totally symmetric coupling coefficient. Suppose we want to consider a scattering process $1\ 2 \rightarrow 1'2'$ to second order in this interaction. Each of the two vertices must have two of the four external lines attached to it. (The only other possibility is that one of the external lines is attached to one vertex and three to the other vertex, but the vertex with three external lines attached to it would have no remaining lines to connect it to the other vertex so this would be a disconnected contribution.) The additional line required at each vertex must then just serve to connect the two vertices to each other. There are three graphs of this type, differing in whether the other external line that is attached to the same vertex as line 1 is line 2 or 1' or 2'. (See Figure 6.8.) Following the rules given above, the contribution to the S -matrix from those three